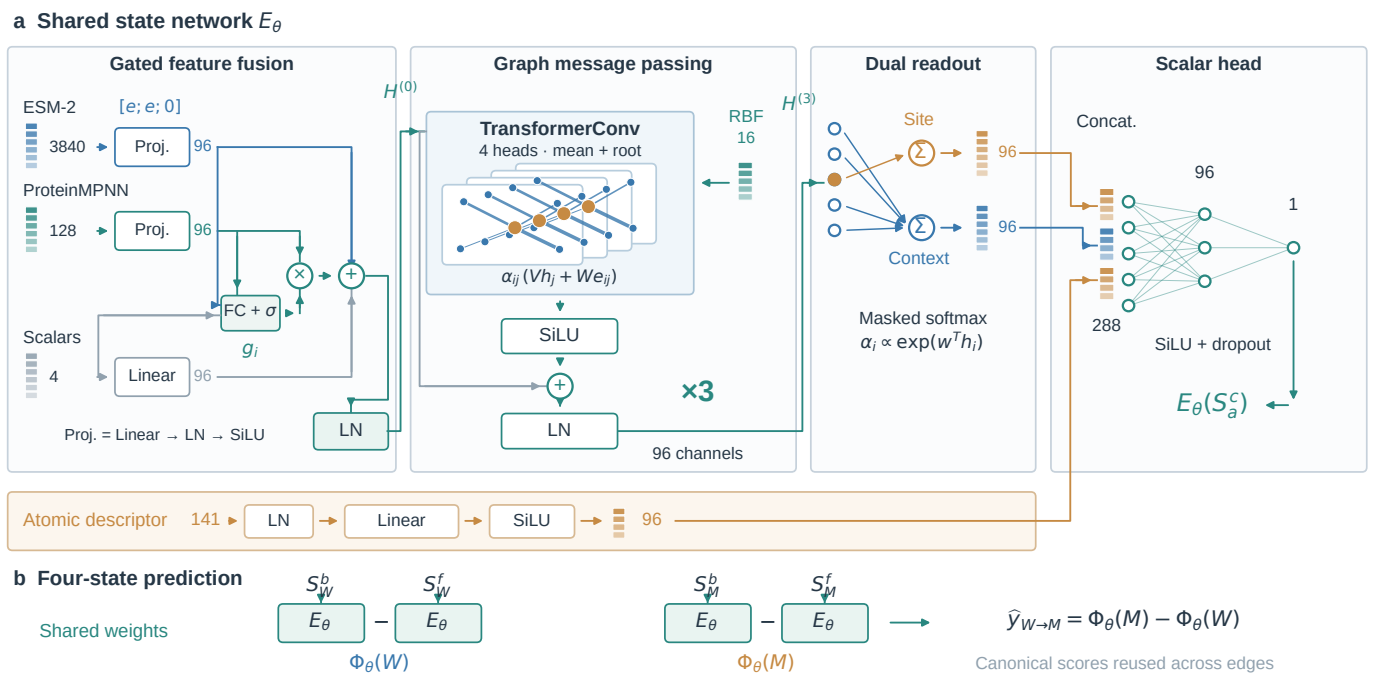

a Shared state network E_θ
Gated feature fusion
ESM-2
[e; e; 0]
3840
Proj.
96
ProteinMPNN
128
Proj.
96
FC + σ
g_i
Scalars
4
Linear
96
Proj. = Linear → LN → SiLU
LN
$H^{(0)}$
Graph message passing
TransformerConv
4 heads · mean + root
$\alpha_{ij}\,(Vh_j + We_{ij})$
RBF
16
SiLU
LN
×3
96 channels
$H^{(3)}$
Dual readout
Site
Context
96
96
Masked softmax
$\alpha_i \propto \exp(w^T h_i)$
Scalar head
Concat.
96
288
1
SiLU + dropout
$E_\theta(S_a^c)$
Atomic descriptor
141
LN
Linear
SiLU
96
b Four-state prediction
Shared weights
S_W^b
S_W^f
E_θ
E_θ
$\Phi_\theta(W)$
S_M^b
S_M^f
E_θ
E_θ
$\Phi_\theta(M)$
$\hat{y}_{W\to M} = \Phi_\theta(M) - \Phi_\theta(W)$
Canonical scores reused across edges